

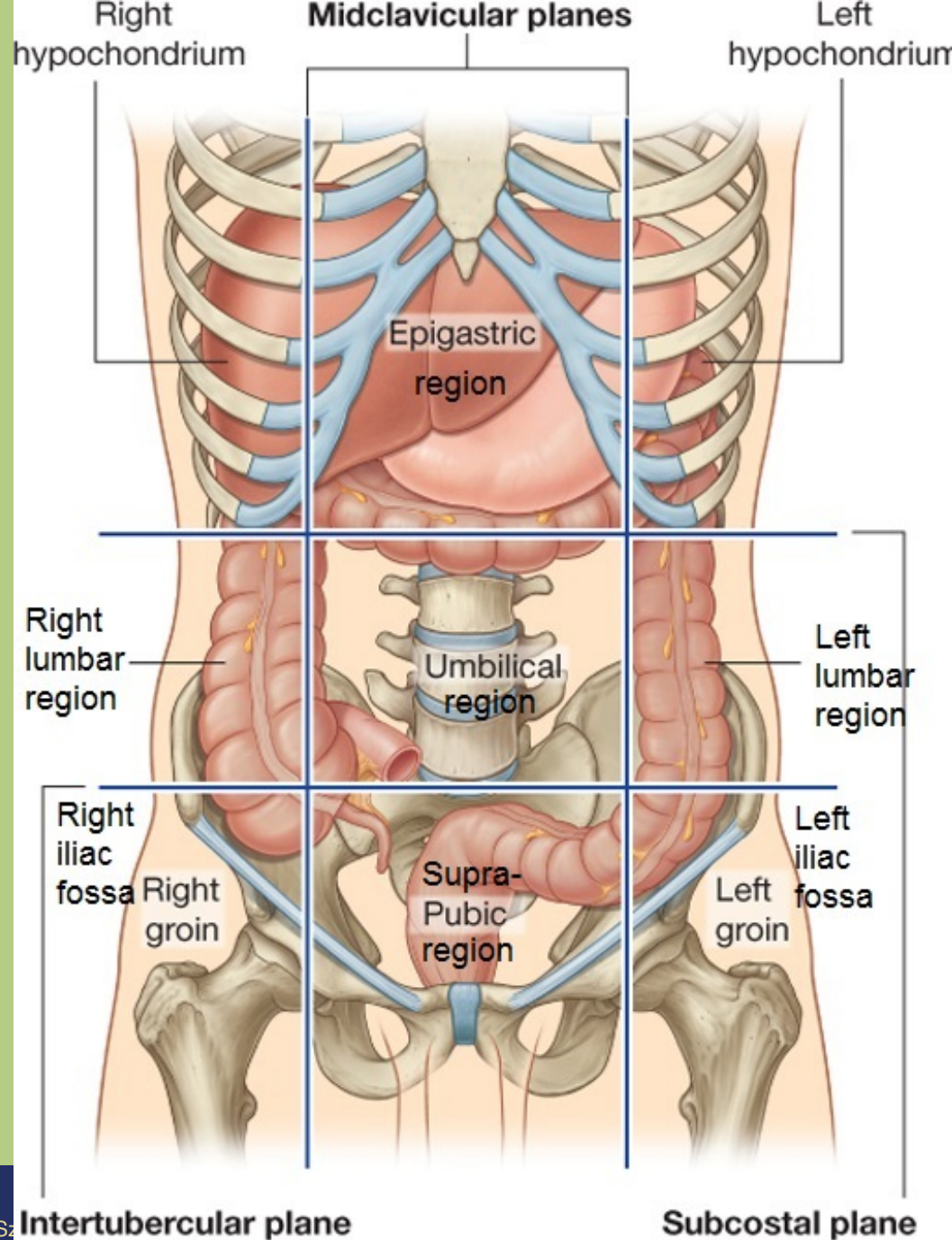
Liver - practice

Dr. Katz Sándor Ph.D.

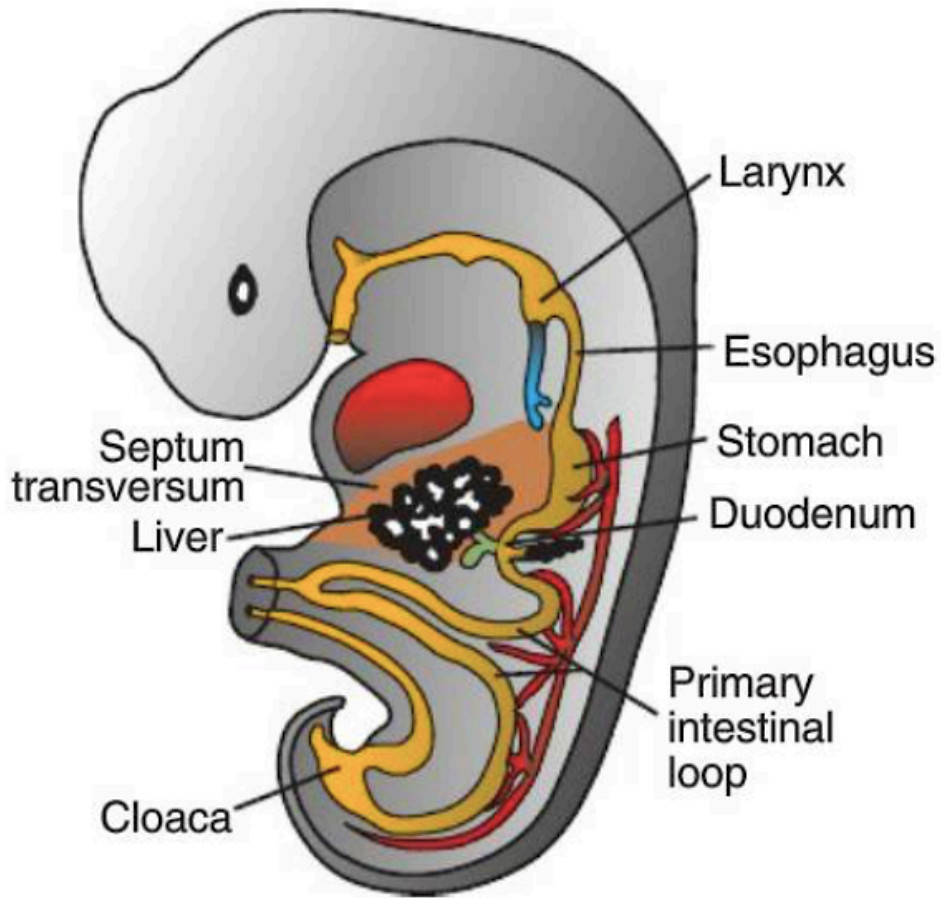


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Location



Ligaments

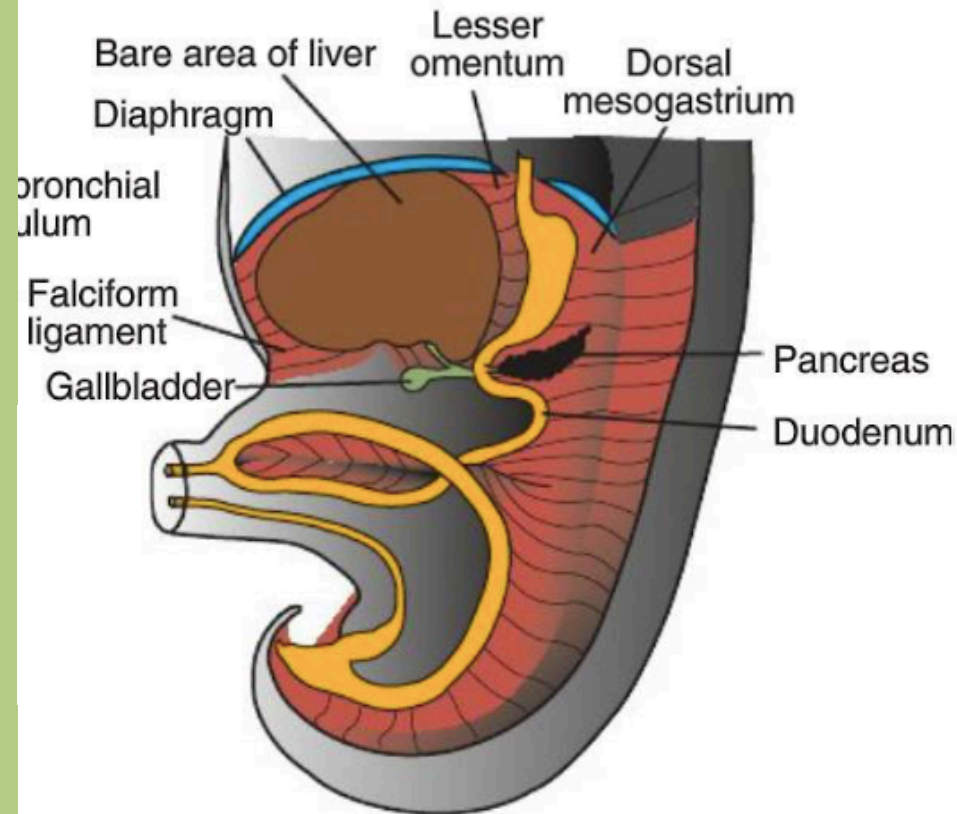


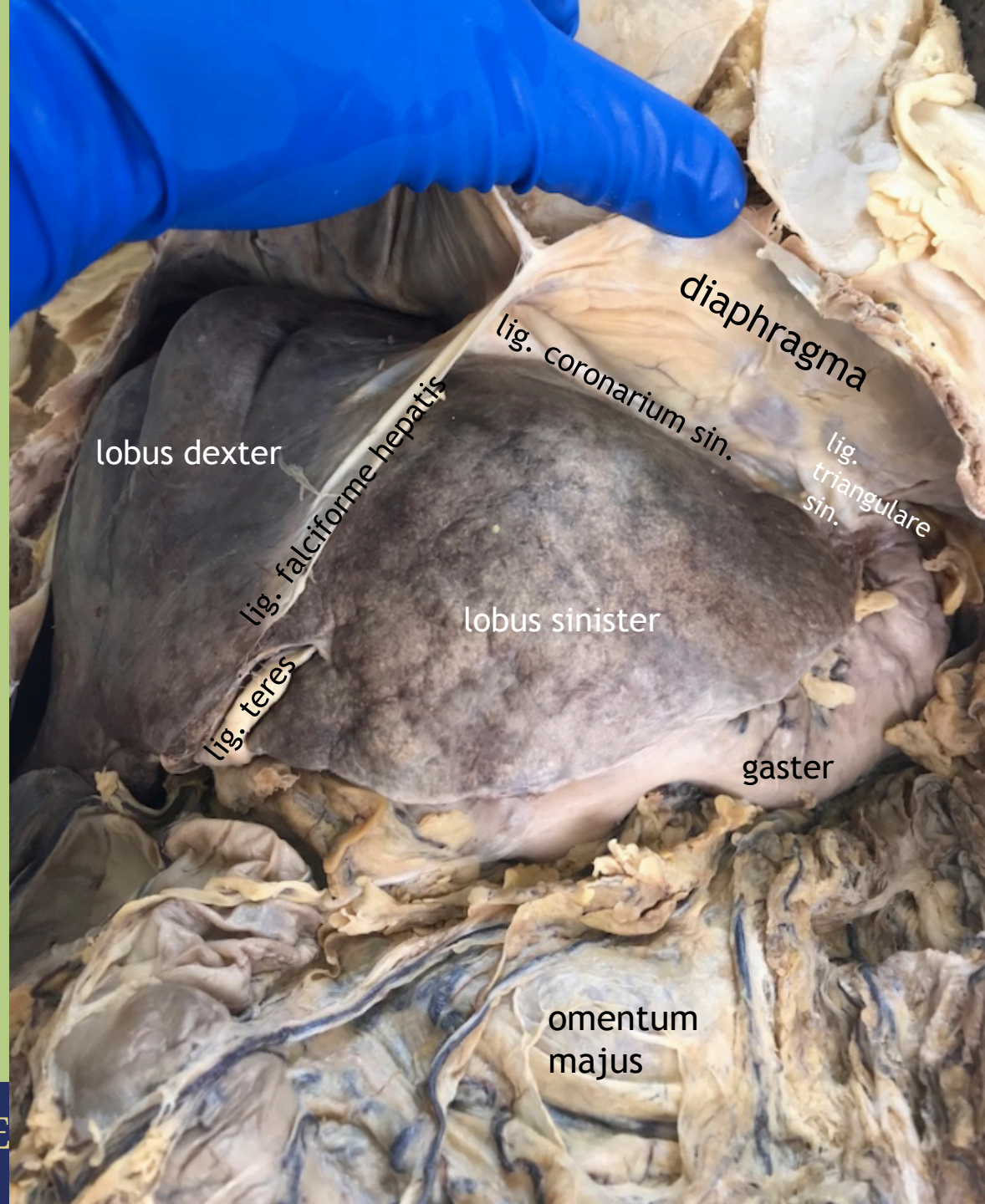
The liver bud is formed by the endodermal proliferation of the foregut, which grows into the septum transversum (a sheet of mesoderm between the pericardium and the yolk sac).

Ligaments

The liver eventually grows into the entire septum transversum and between the anterior abdominal wall and the liver, and between the foregut and the liver, ligaments are formed: falciform lig. and the lesser omentum.

The mesoderm forms the visceral peritoneum on the surface of the liver, except on the cranial surface (area nuda=bare area), where it is directly attached to the septum transversum. This will later become the centrum tendineum of the diaphragm.





lobus dexter

lig. falciforme hepatis

lig. teres

lobus sinister

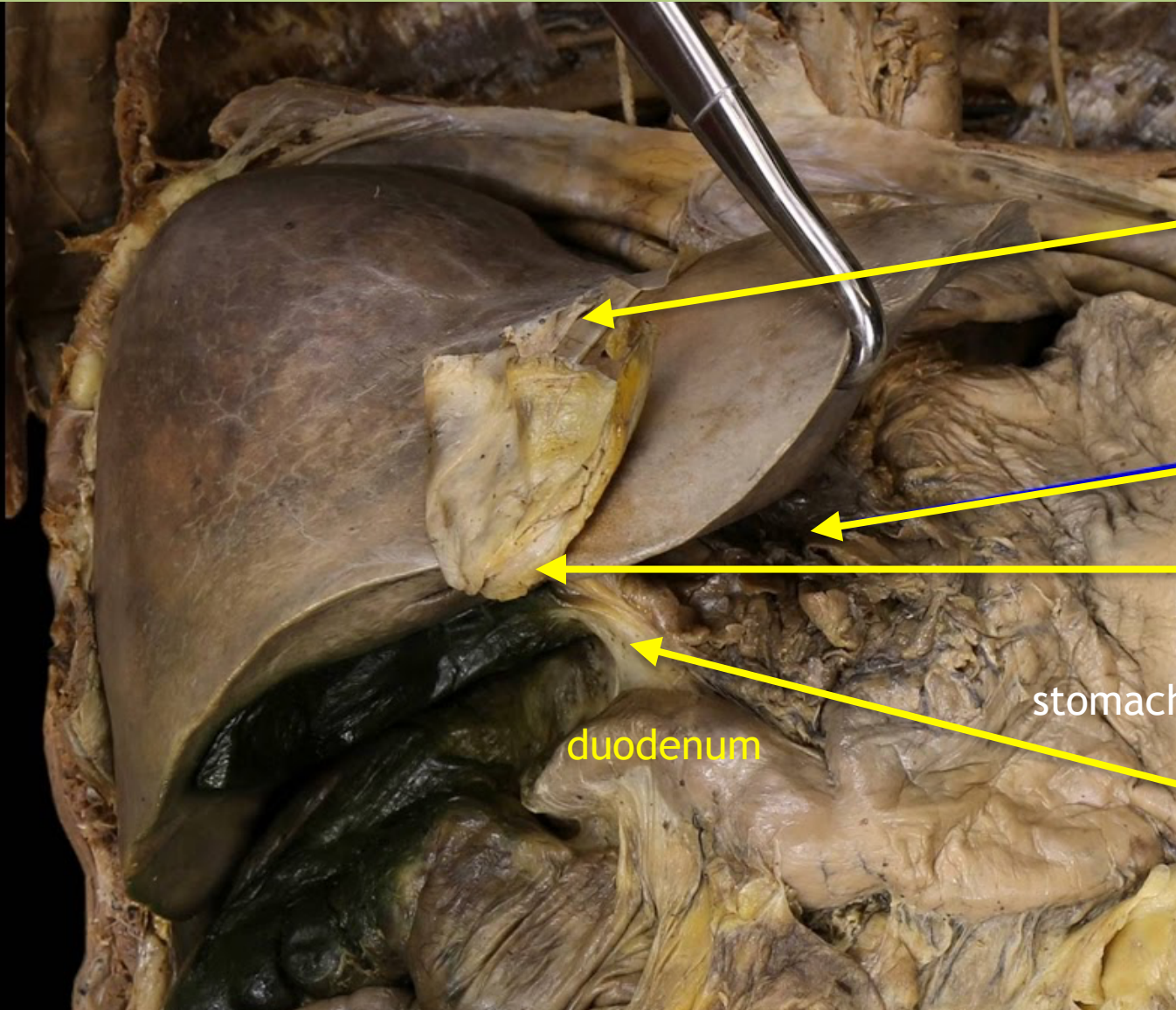
lig. coronarium sin.

diaphragma

lig.
triangulare
sin.

gaster

omentum
majus



Lig. falciforme
hepatis

Lig. hepatogastricum
(omentum minus)

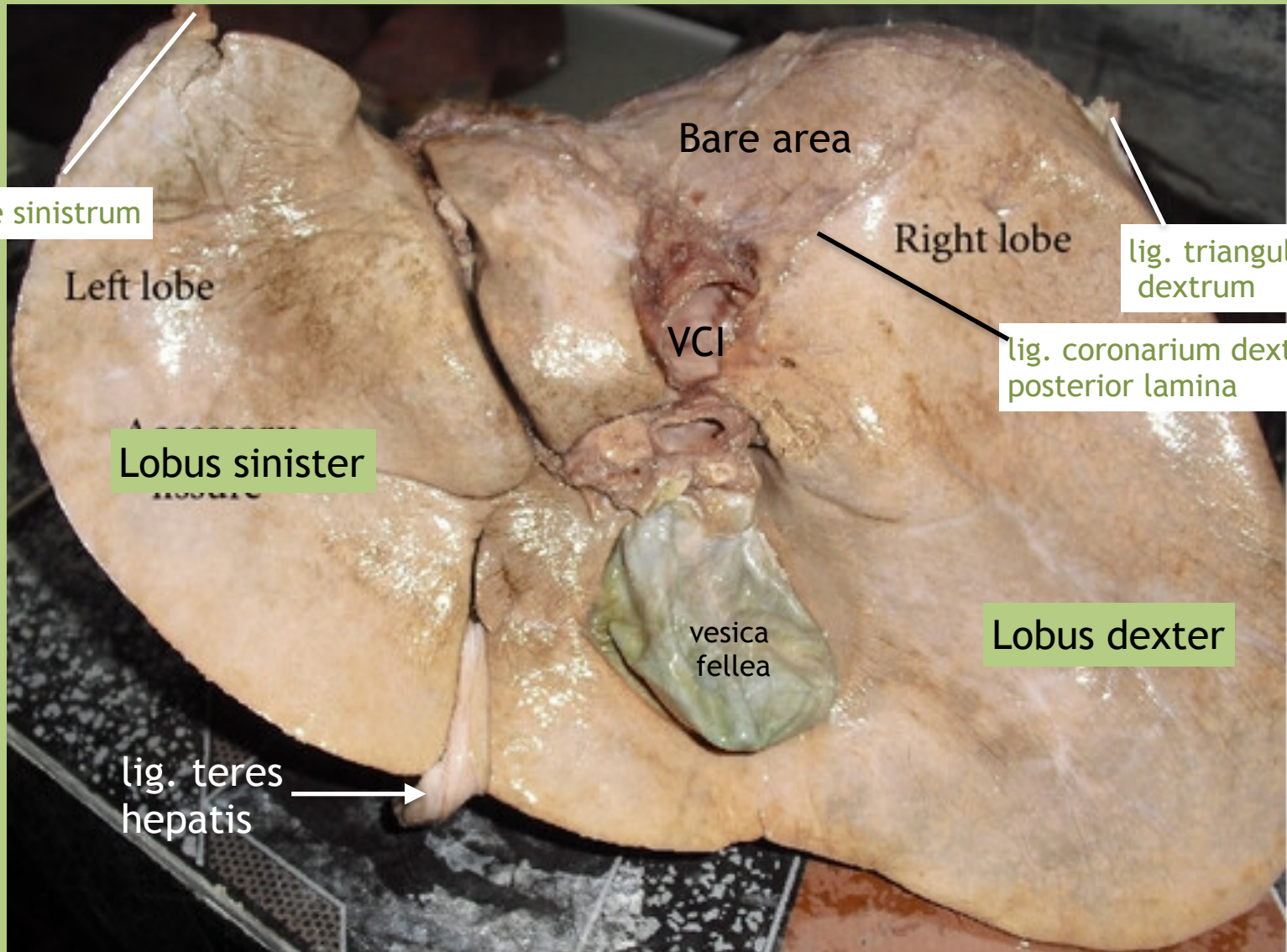
Lig. teres
hepatis (left
umbilical v.)

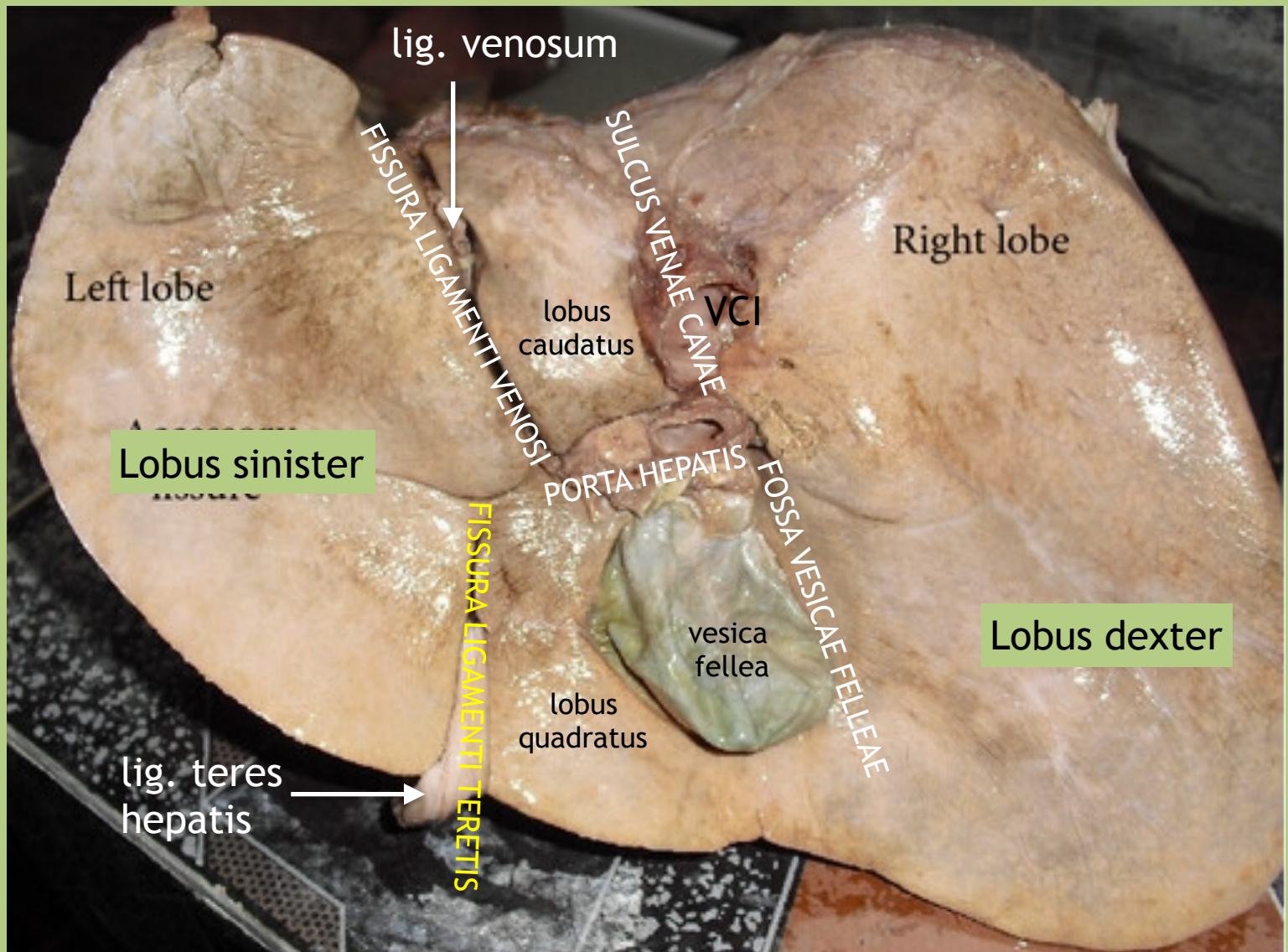
stomach

duodenum

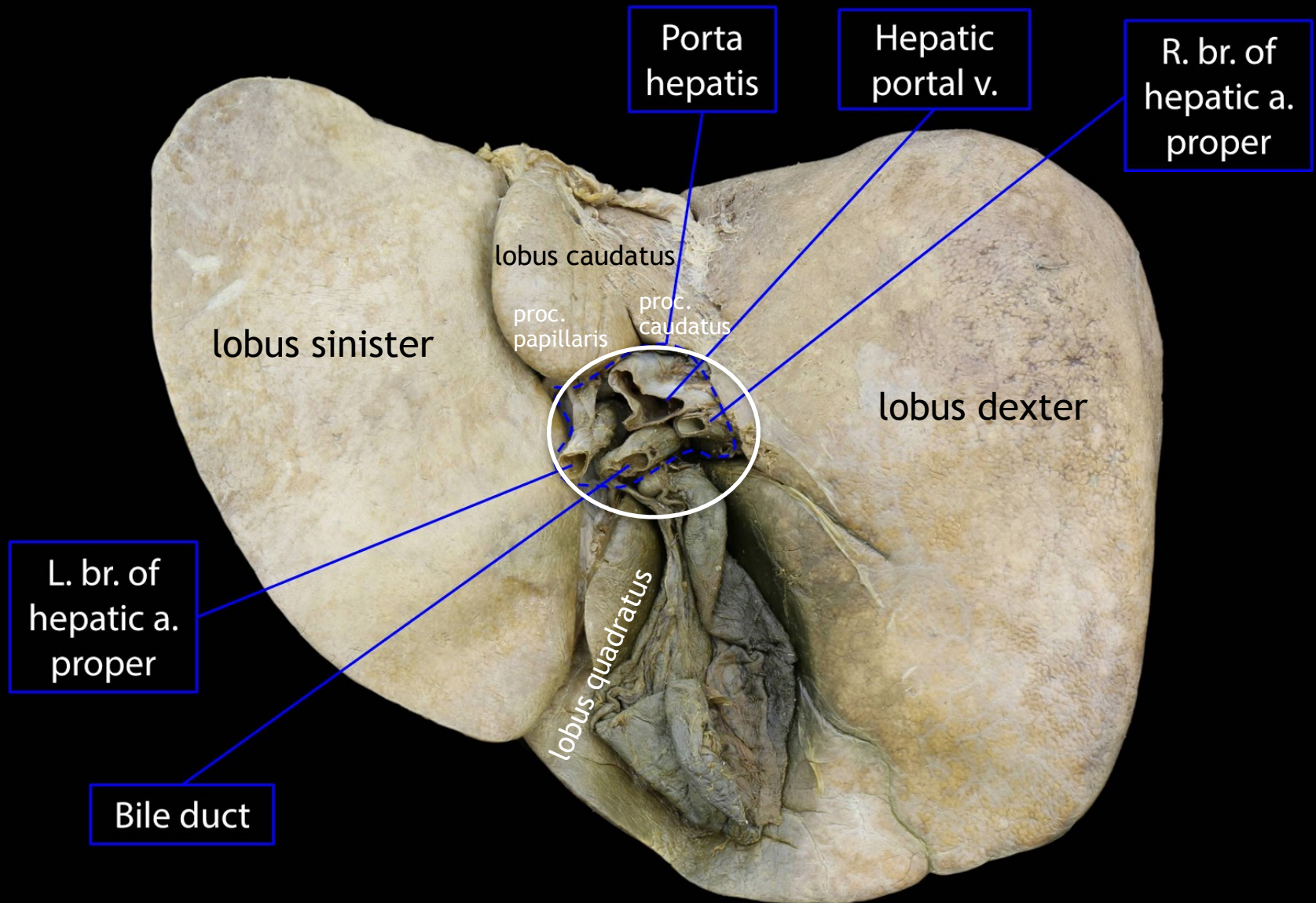
Lig.
hepatoduodenale
(omentum minus)

Facies visceralis - Ligamenta

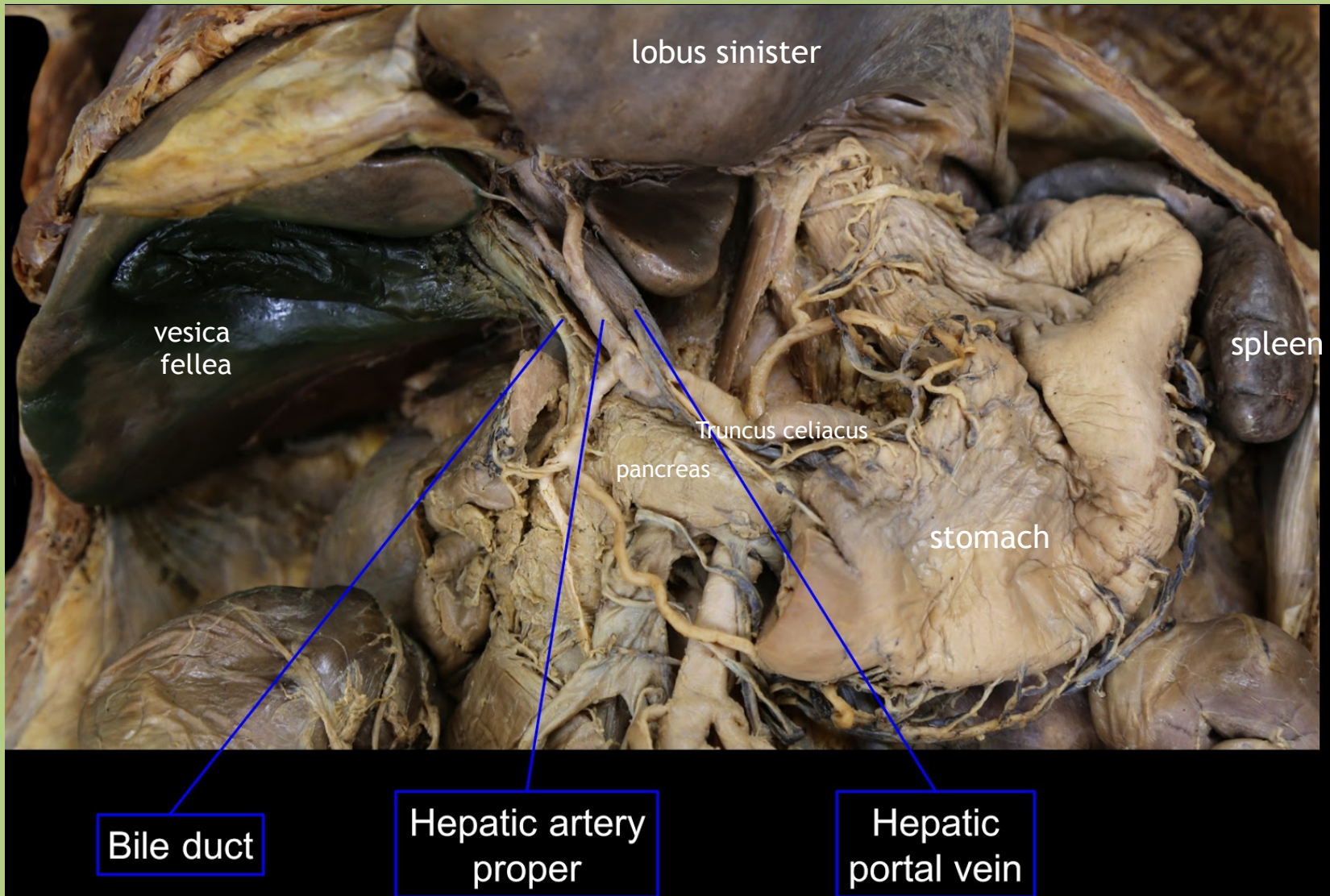




Porta hepatis

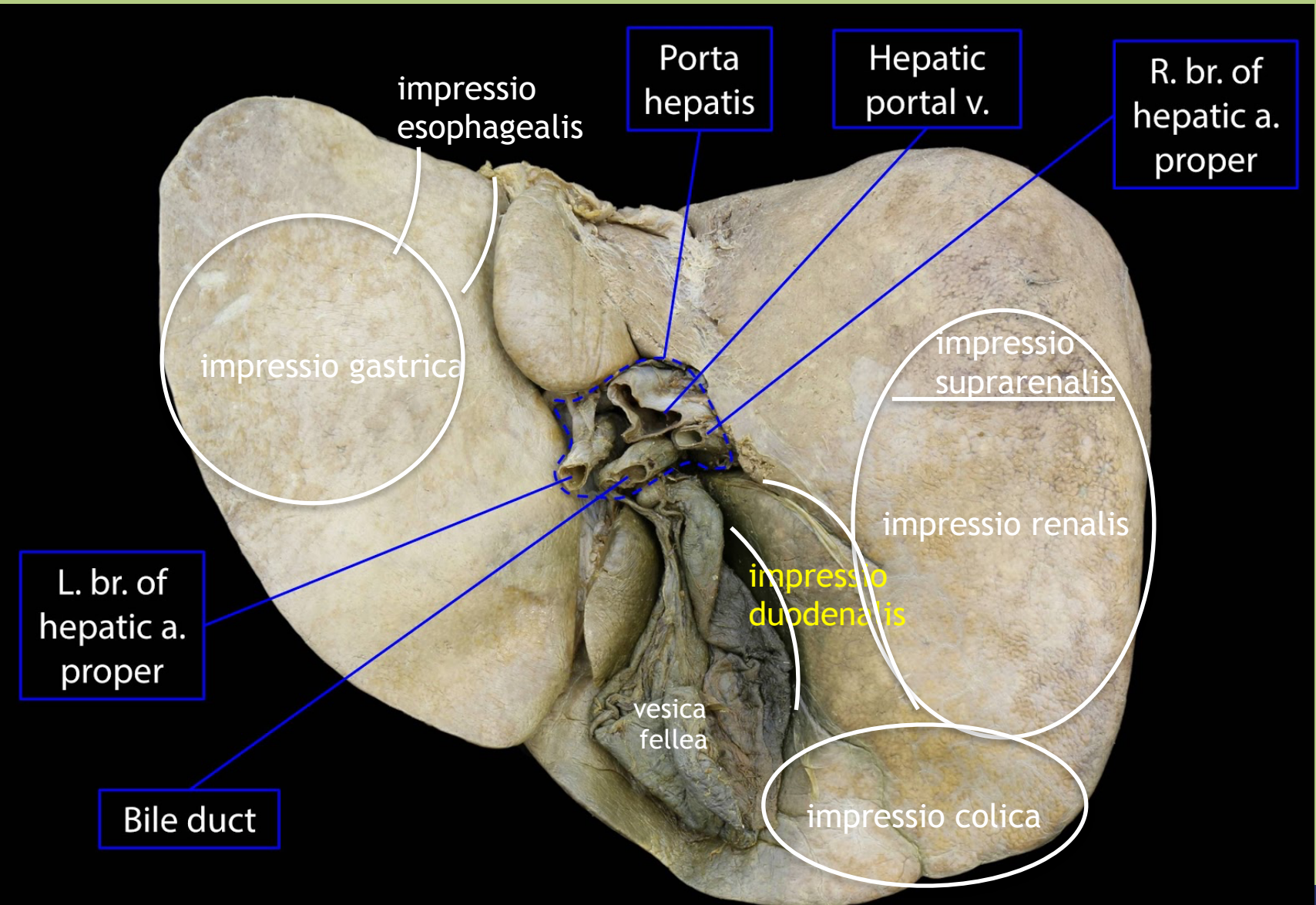


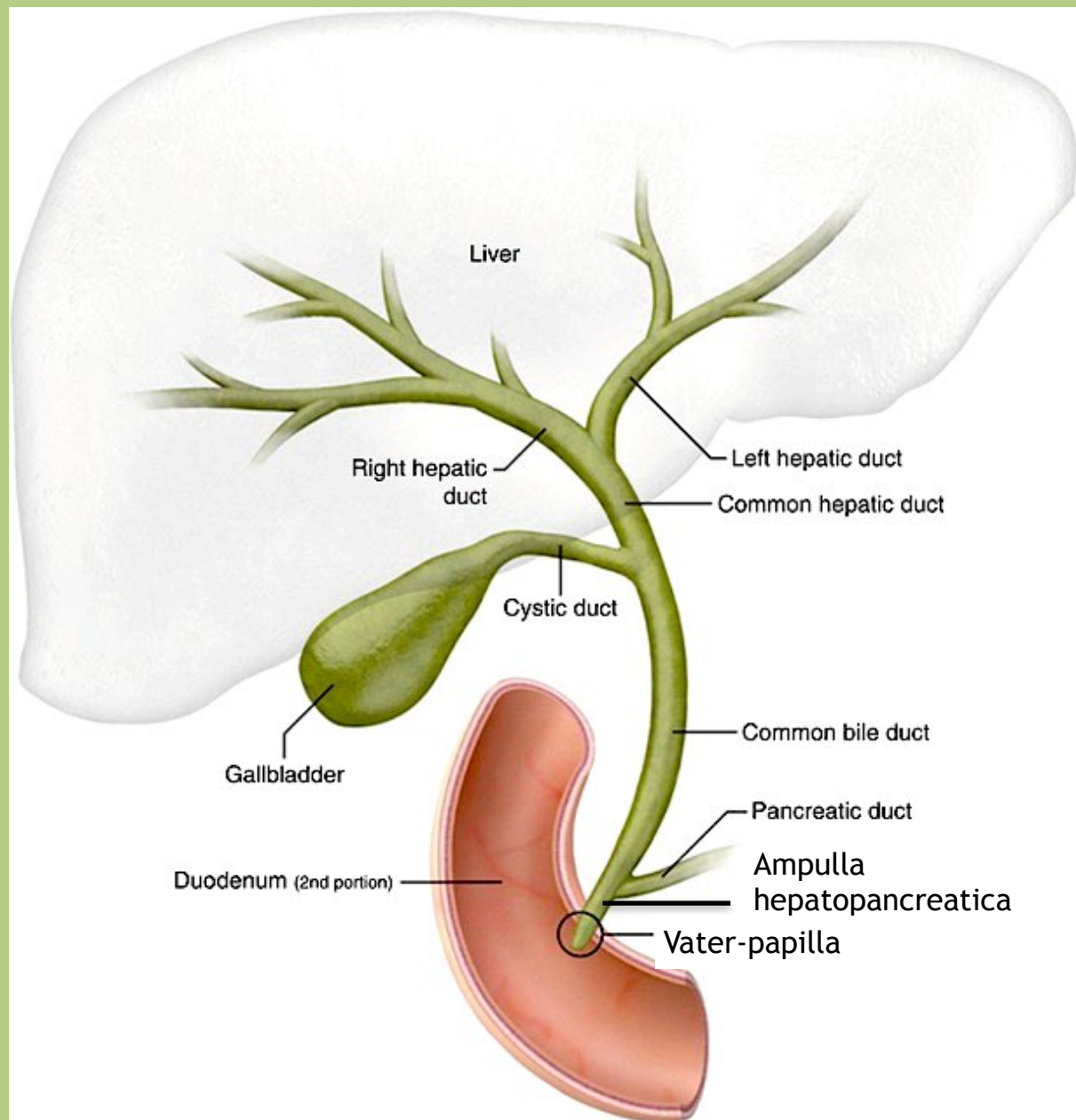
Porta hepatis

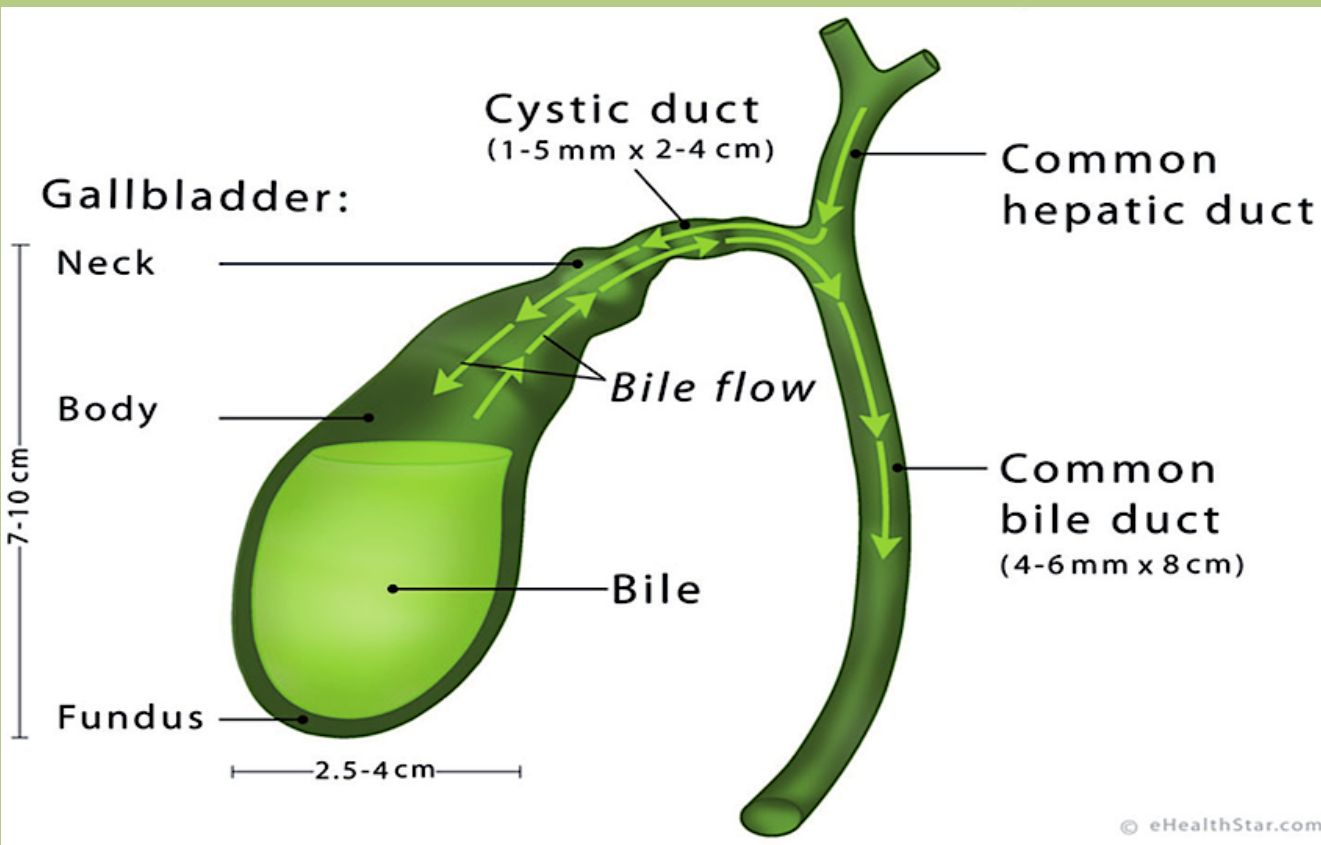


Hepatoduodenal lig. contents, anterior

Impressions

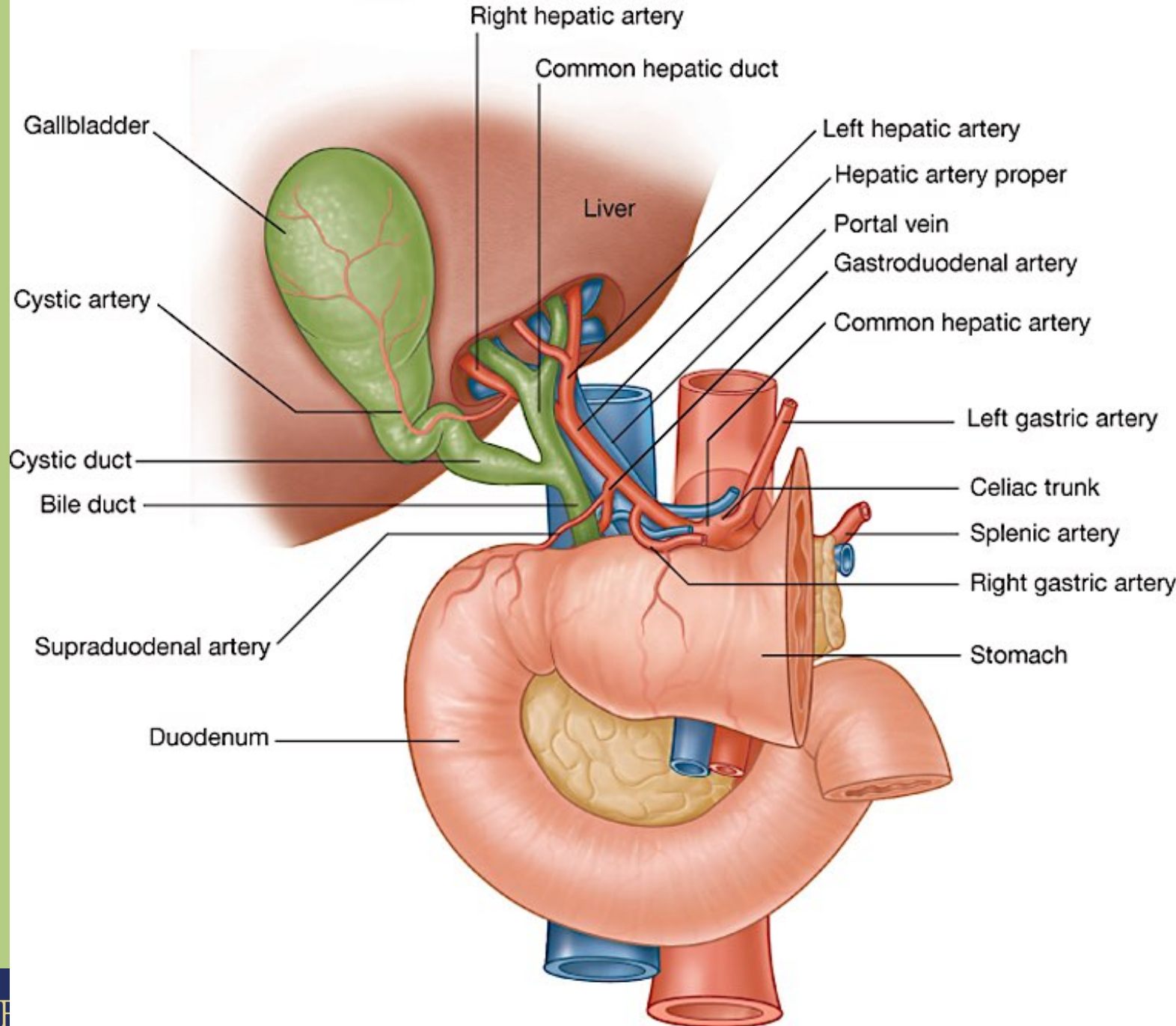






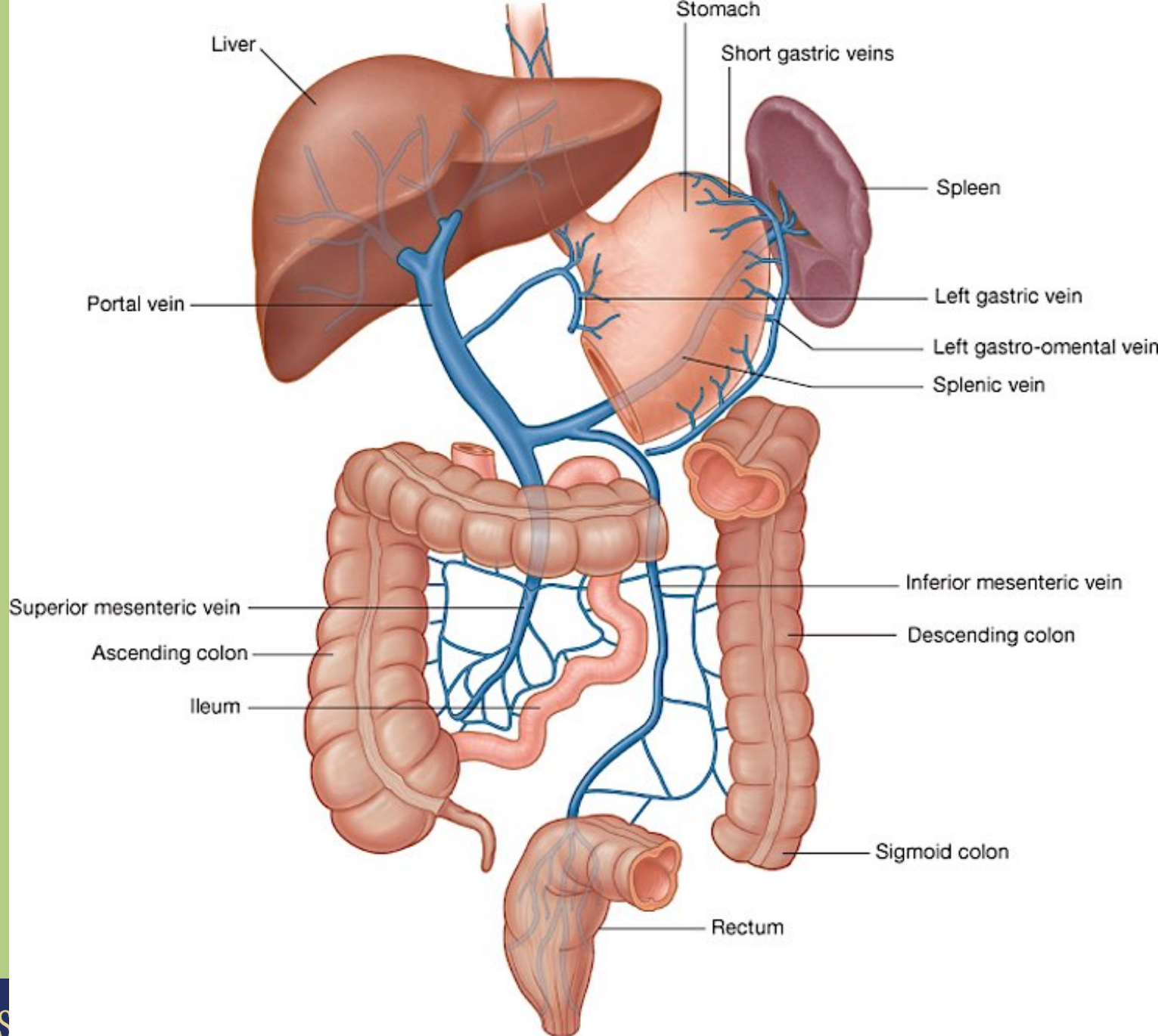
Vasa privata:

a. hepatica propria



Vasa publica:

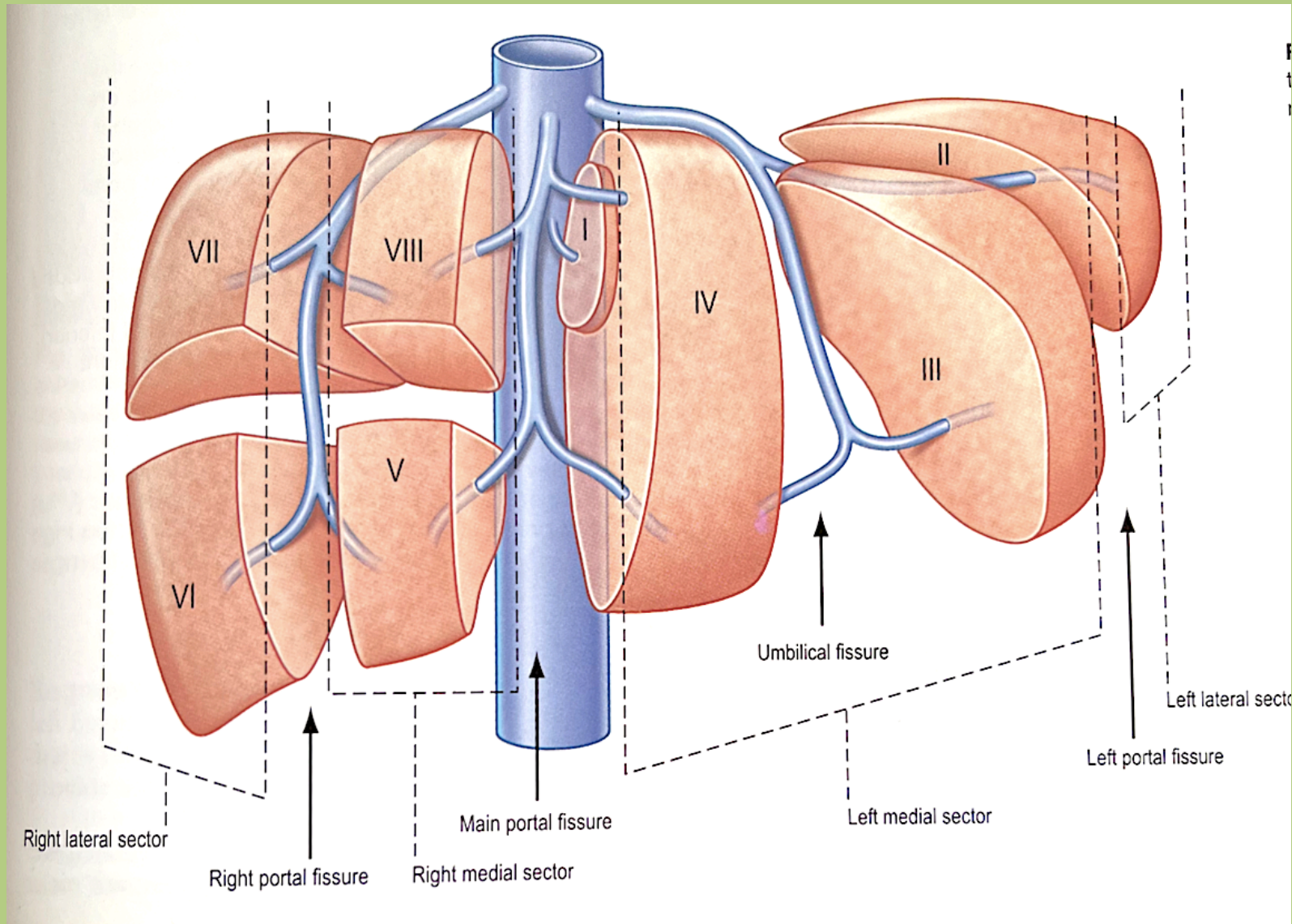
v. portae



Segments

COMMON VENOUS DRAINAGE

hepatic
veins



PORTO-CAVAL ANASTOMOSES

In a healthy body, the venous connections between the two systems are closed or insignificant.

However, if liver disease (cirrhosis) causes increased pressure in the portal system (portal hypertension), these anastomoses enlarge, dilate, and rupture, causing severe bleeding.

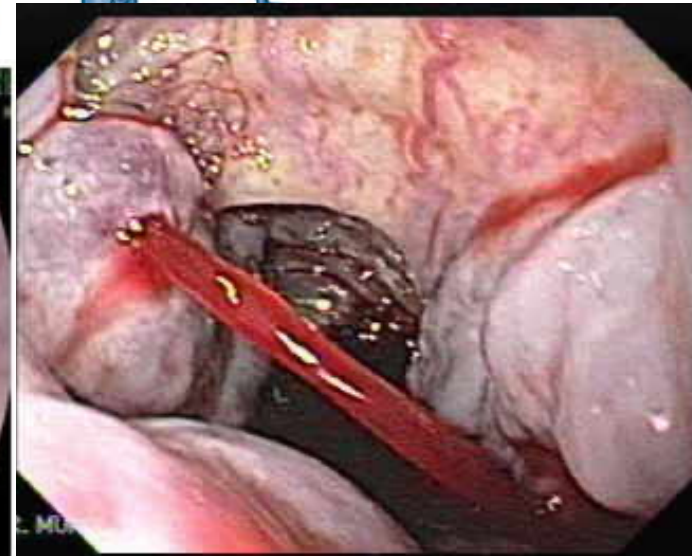
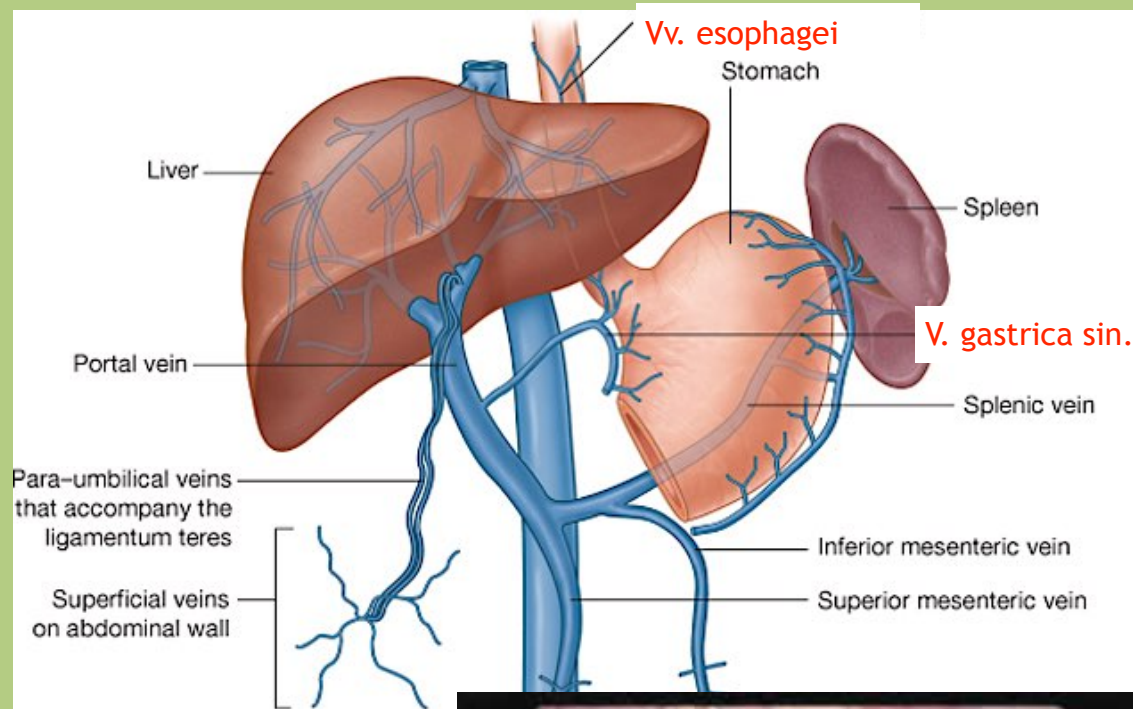


PORTO-CAVAL ANASTOMOSIS I.

Vein belonging to the portal vein system: **left gastric vein**

Vein belonging to the v. cava system: **esophageal veins** (leading to the azygos system)

Clinical appearance: dilated veins in the esophagus and upper part of the stomach. Possibility of life-threatening bleeding.

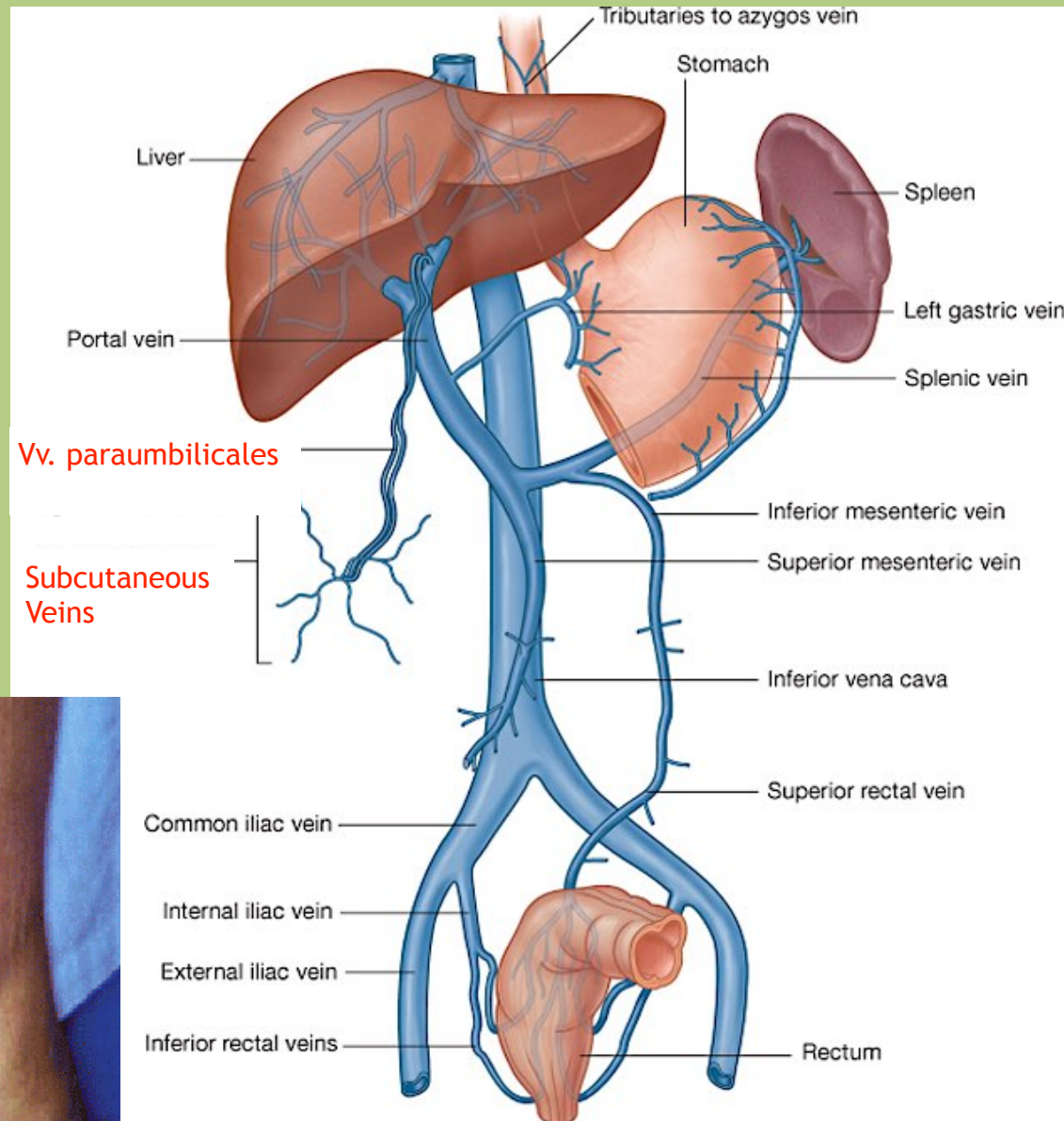


PORTO-CAVAL ANASTOMOSIS II.

Vein belonging to the portal vein system: **paraumbilical veins** (thin branches of the left branch of the portal vein remaining in the lig. teres hepatis)

Vein belonging to the v. cava system: **subcutaneous veins towards the superior and inferior epigastric veins**

Clinical appearance: caput medusae

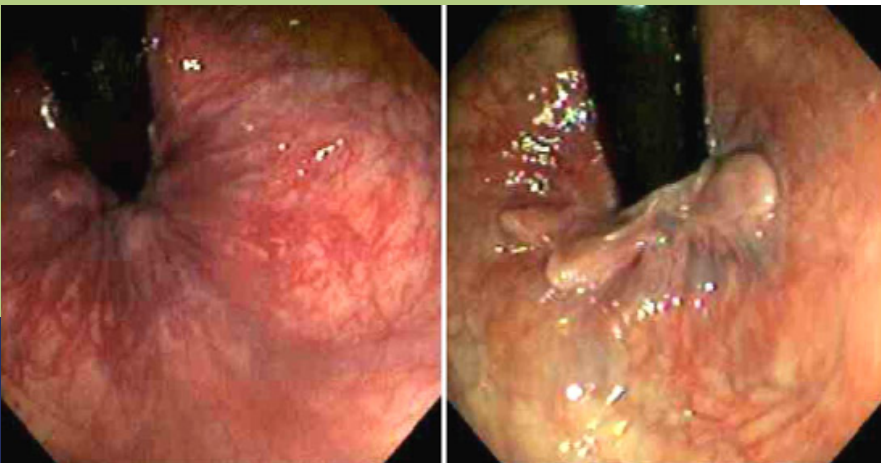
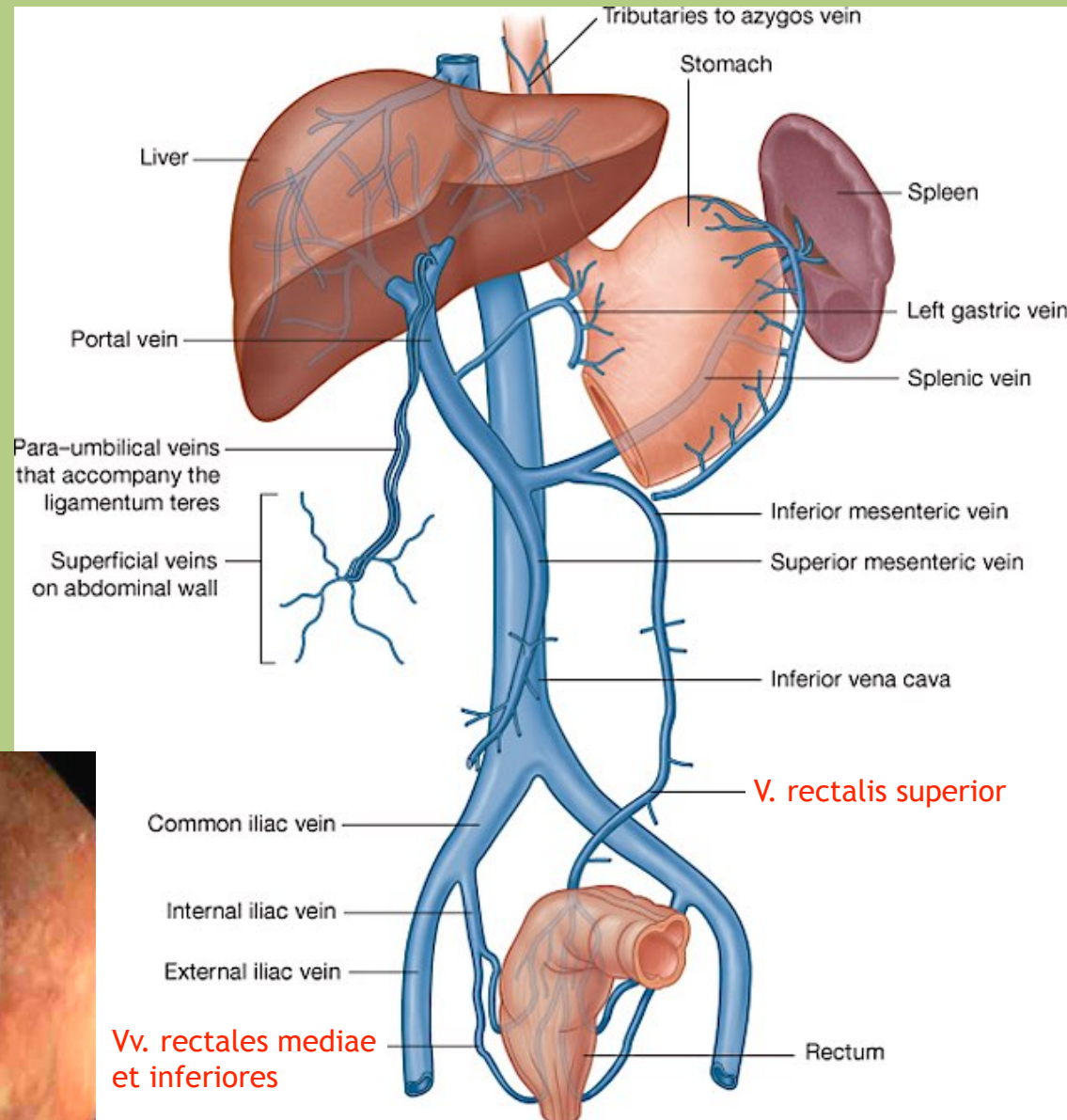


PORTO-CAVAL ANASTOMOSIS III.

Vein belonging to the portal vein system: **superior rectal vein**

Vein belonging to the v. cava system: **middle and inferior rectal veins** towards the internal iliac vein

Clinical appearance: hemorrhoidal nodes

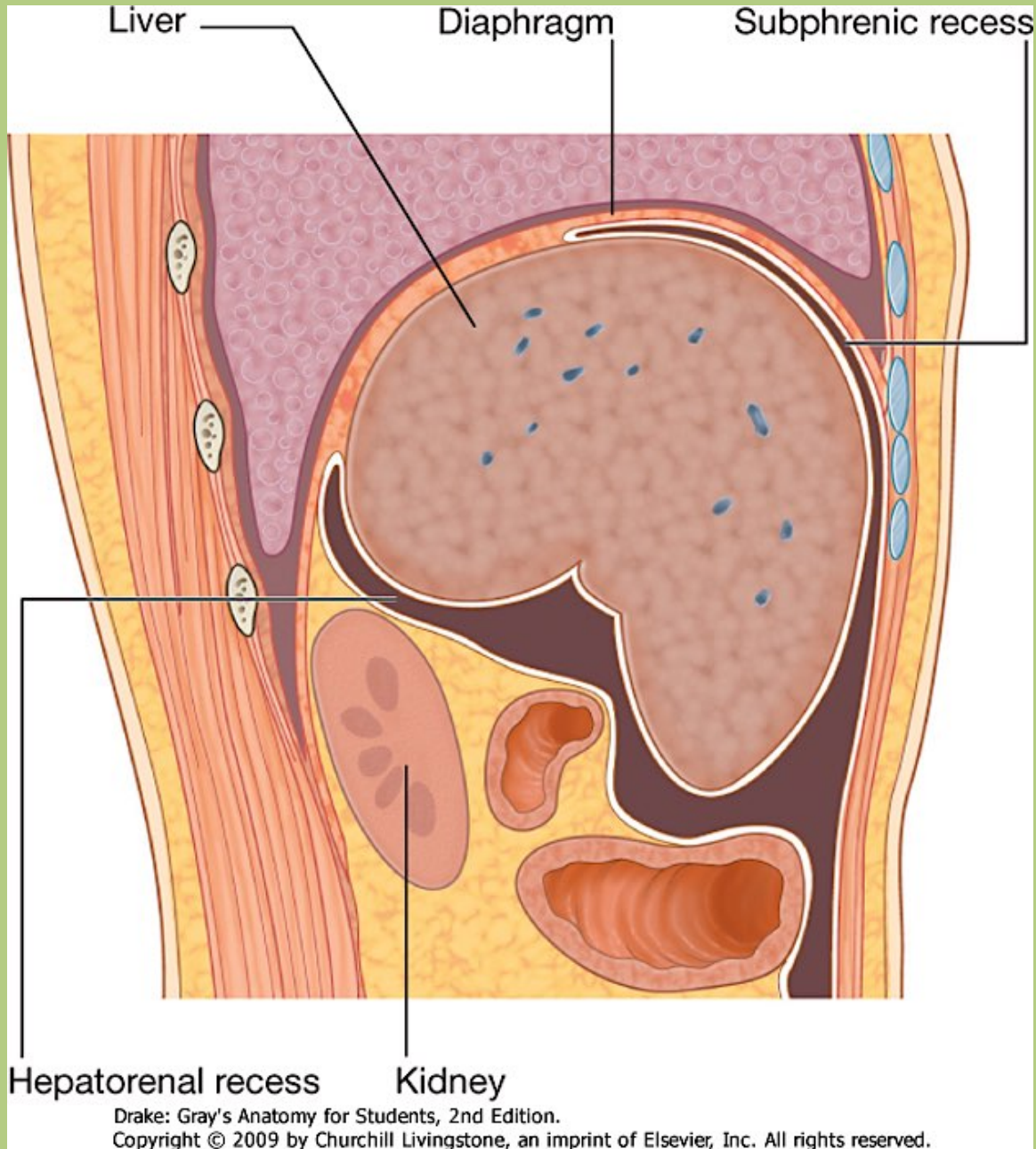


PERITONEAL AREAS AROUND THE LIVER

Subphrenic recess: between the lower surface of the diaphragm and the liver

Hepatorenal recess (Morison's pouch): between the liver and the right kidney; deepest point of the abdominal cavity in the supine position; communicates with the cavity of the omental bursa (behind the stomach) through the epiploic foramen (Winslow).

Clinical importance: fluid accumulation, space for abscess or metastasis.



Köszönöm a figyelmet!

References:

Gray's Anatomy for Students

Standing: Gray's Anatomy

Thieme, Atlas of Anatomy, Neck and Internal Organs

LWW: Langman's Medical Embryology

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